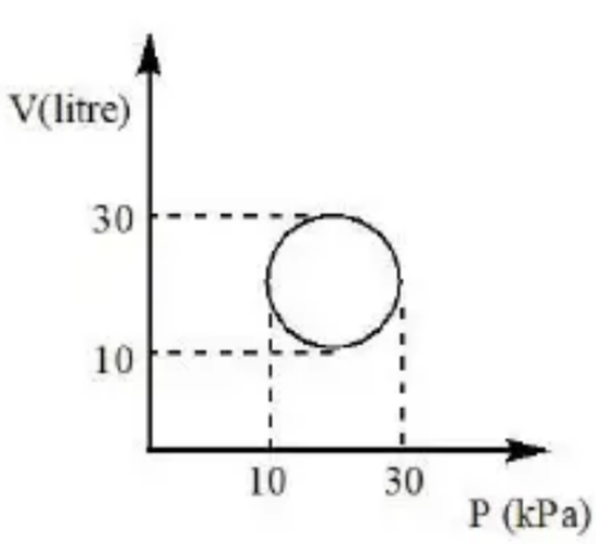
2 5

4 3

4 3



10^7 J

10^4 J

10^2 J

10^3 J

λ ρ λ

,

$\frac{1}{2}, 0, \frac{1}{2}$

1, 1, 0

$\frac{1}{2}, \frac{1}{2}, 0$

$40\,\mathrm{kg\,m^2}$

4^0

0^0

4^0

$\frac{1}{3}$ -2

1.2 10^{5} 1

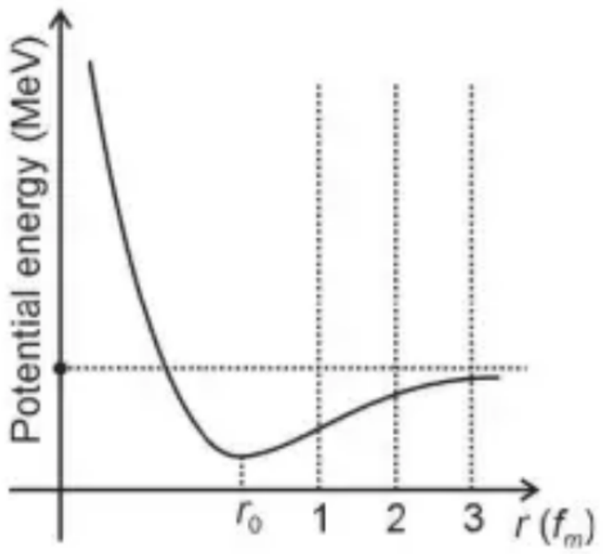
$\sqrt{3}$

$\frac{5}{2}$

$\frac{3}{2}$

10 2

$$\frac{-}{65}\ \frac{-}{},$$



$\frac{1}{2}$

$\frac{1}{4}$

$\frac{1}{\sqrt{2}}$

0.4

Section-B

186. T

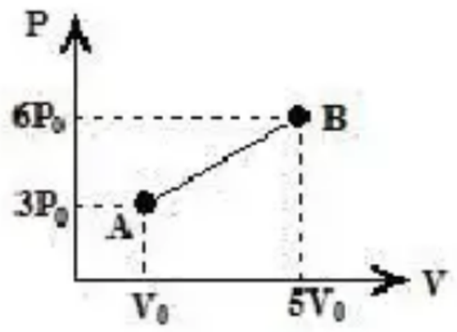
$\vec{r} = 3\hat{} + 2^{2}\hat{} + 5\hat{}$ '*t*' is in seconds and the coefficients have proper units for '*r*' to be in metre. The velocity of particle at t =1s is ($\sin 37^0 = 0.6$)

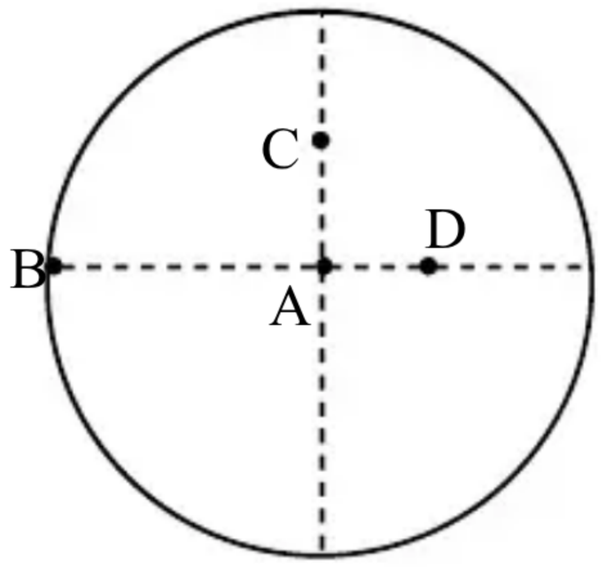
(1) $5\ ms^{-1}, 37^0$ with x-axis

(2) $5\ ms^{-1}, 53^0$ with x-axis

(3) $4\ ms^{-1}, 37^0$ with x-axis

(4) $4\ ms^{-1}, 53^0$ with x-axis





(1) B (2) C

(3) D (4) A

(1) $\frac{}{6}$ (2) $\frac{}{4}$

(3) $\frac{}{9}$ (4) $\frac{2}{9}$

193. Prasad peddles a stationary bicycle peddles attached to a 100 turn coil of area 0.1 m^2. The coil rotates at half revolution per second. If a uniform magnetic field of induction 0.01T acts perpendicular to the axis of rotation of coil, then the maximum voltage generated in the coil is

(1) 0.314 V

(2) 1.256 V

(3) 0.157 V

(4) 0.628 V

194. A sinusoidal voltage of peak value 283 V and frequency 50 Hz is applied to a series LCR circuit in which $R = 3\ \Omega$, capacitive reactance, $X_C = 4\ \Omega$ and inductive reactance, $X_L = 8\ \Omega$. The power dissipated in the circuit is

(1) 4.8 kW

(2) 2.4 kW

(3) 3.6 kW

(4) zero

195. A small bulb is placed at the bottom of tank containing a liquid to a depth of 30 cm. The area of surface of liquid, through which light from bulb can average out is (Refractive index of liquid is 5/4)

(1) 5026 cm^2 (2) 8044 cm^2

(3) 6025 cm^2 (4) 3256 cm^2

196. Which among the following is not a Maxwell's equation of electromagnetism? (Symbols have their usual meaning)

(1) $\oint \vec{E} \cdot d\vec{A} = \frac{q}{\varepsilon_0}$

(2) $\oint \vec{B} \cdot d\vec{A} = 0$

(3) $\oint \vec{B} \cdot \vec{dl} = \mu_0 \left(I_C + \varepsilon_0 \frac{d\phi_E}{dt} \right)$

(4) $\oint \vec{E} \cdot \vec{dl} = \varepsilon_0 q + \frac{d\phi_B}{dt}$

197. Two electric bulbs rated P_1 watt V volts and P_2 watt V volts are connected in series and V volts are applied to it. The total power will be

(1) $P_1 + P_2$ *watt*

(2) $\sqrt{P_1 P_2}$ *watt*

(3) $\frac{\sqrt{P_1 P_2}}{P_1 + P_2}$ *watt*

(4) $\frac{P_1 P_2}{P_1 + P_2}$ *watt*

198. α – particle consists of

(1) 2 electrons, 2 protons and 2 neutrons

(2) 2 electrons and 4 protons only

(3) 2 protons only

(4) 2 protons and 2 neutrons only

199. A telescope has an objective and eye piece of focal lengths 144 cm and 6 cm. The magnifying power of telescope is

(1) 48 (2) 150

(3) 24 (4) 42

200. Match Column I with Column II.

	Column I		**Column II**
A)	Carburetor of vehical	P)	Lorentz force
B)	Hydraulic brakes	Q)	Eddy currents
C)	Mass spectrometer	R)	Bernoulli's principle
D)	Magnetic brakes	S)	Pascal's law

	A	**B**	**C**	**D**
(1)	R	S	P	Q
(2)	S	R	P	Q
(3)	P	S	R	Q
(4)	R	P	S	Q